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METHOD FOR CONTINUOUS MONITORING AND OPTIMIZATION OF RECOVERY PROCESS

Abstract

Systems and methods for exacting elements of interest, such as lithium, from an aqueous material are described herein. The systems generally use multiple vessels with selective media in each vessel to accomplish extraction of the element of interest. The vessels are operated in cyclic, permuted fashion to move between absorption and desorption operations by routing flows of streams for accomplishing such operations among the vessels in programmed ways. Sensors configured to detect total dissolved solids, or a parameter related to total dissolved solids, are used to detect endpoints for mode switching and to ascertain other aspects of system performance.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This patent application claims benefit of U.S. Provisional Patent Application Ser. No. 63/562,299 filed Mar. 7, 2024, which is entirely incorporated herein by reference.

FIELD

[0002] This patent application describes methods and apparatus for recovery of an element of interest, such as lithium, from aqueous sources. Specifically, processes and apparatus for monitoring a recovery process are described.

BACKGROUND

[0003] A number of ions can be sourced from aqueous materials (or brines), such as aqueous material present at or near the surface of the earth. Ions such as lithium, manganese, nickel, cobalt, and others can be extracted using direct aqueous extraction. Aqueous materials subjected to such extraction can have different compositions that include various critical minerals-derived ions, as well as different contaminants.

[0004] For instance, lithium is a key element used in batteries for electric vehicles, portable electronics, and renewable energy storage systems. For example, electrical storage devices, such as batteries, supercapacitors, and other devices commonly use lithium to mediate the storage and release of chemical potential energy as electrical current. Other critical minerals may be used in batteries and/or other devices related to renewables energy for example. As the demand for renewable, but non-transportable, energy sources such as solar and wind energy grows, so does the demand for technologies to store energy generated using such sources also grows and, in turn, the demand for lithium also grows.

[0005] The mining industry has numerous techniques for the extraction of critical minerals, such as lithium. Hard rock mining with acid digestion is common, but labor and energy intensive. Another prevalent technique involves utilizing surface waters and salar lakes. This method comprises pumping brine into evaporation ponds with the addition of chemical agents to selectively precipitate lithium, the brine is evaporated (e.g., by solar energy) leaving behind concentrated lithium salts. Lithium extraction by evaporation techniques may be slow, environmentally disruptive (e.g., due to water usage, land disruptions, and potential leakage of toxic chemicals into surrounding areas), and dependent on climate conditions. For example, lithium extraction with evaporation ponds may take months (e.g., up to 18 months) to complete, recovering roughly 50-60% of the original lithium.

[0006] Direct extraction of critical minerals, such as Direct Lithium Extraction (DLE) is a set of advanced technologies designed to extract critical minerals from brine sources, such as salt flats or underground aquifers. In this application, a brine source or brine is defined as an aqueous source containing the dissolved element of interest, including ions derived from the element of interest. Such brine may originate from natural or artificial source and can include tailings, wastewater, battery recycling, oilfield stream, seawater, hard rock leachate, etc. The element of interest may include more specifically one or more of lithium, nickel, cobalt, manganese, magnesium, potassium, copper, iron, zinc, aluminum, molybdenum, or vanadium.

[0007] Direct extraction aims to accelerate the extraction process while avoiding the need for extensive evaporation and minimizing environmental impact. Rich brines may be found for instance in regions with high geological activity, such as salt flats (salar) and underground aquifers.

[0008] Direct extraction offers several benefits over traditional methods. Direct extraction methods can significantly reduce the time required to extract the element of interest from brine sources, potentially down to a matter of hours. Direct extraction technologies can be designed to require less

space compared to expansive evaporation ponds. Direct extraction technologies have the potential to reduce water usage, a critical consideration in water-scarce regions. By reducing the need for large evaporation ponds and lowering the risk of chemical leakage, direct extraction methods can have a reduced environmental impact. Direct extraction technologies can be easily scaled up or down based on demand, unlike traditional methods, which rely heavily on climate conditions.

SUMMARY

[0009] Embodiments described herein provide a system for extracting an element of interest from an aqueous material, the system comprising a first vessel containing a first selective medium to extract an element of interest from an aqueous material; a second vessel containing a second selective medium to extract the element of interest from the aqueous material; a source of the aqueous material fluidly couplable to the first vessel and the second vessel; an eluent source fluidly couplable to the first vessel and the second vessel; a first sensor to measure total dissolved solids (TDS), or a parameter related to TDS, at the first vessel; a flow system to direct the aqueous material to the first vessel or to the second vessel and to direct the eluent to the first vessel or to the second vessel; and a controller configured to control the flow system to route one of the aqueous material and eluent to the first vessel; and iteratively vary a flow rate through the first vessel, for each iteration control the first sensor to measure TDS, or the parameter related to TDS, at the first vessel, and determine a target value of the flow rate through the first vessel based on one or more of the measurements.

[0010] Other embodiments described herein provide a method, comprising extracting an element of interest from an aqueous material using at least a first vessel containing a first selective medium and a second vessel containing a second selective medium, wherein extracting the element of interest includes coupling a source of the aqueous material to the first vessel and to the second vessel to load the element of interest onto the first and second selective medium and coupling an eluent source to the first vessel and to the second vessel to unload the element of interest from the first and second selective medium, wherein the method further includes routing a target fluid to the first vessel, wherein the target fluid is one of the aqueous material and eluent; iteratively varying the flow rate of the target fluid to the first vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the first vessel for each iteration; and determining a target value of the flow rate of the target fluid through the first vessel based on the one or more of the measurements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a flow diagram summarizing a method of obtaining a substantially continuous reading of a difference in lithium concentration between two aqueous materials according to one embodiment.

[0012] FIG. 2 is a flow diagram summarizing a method of determining lithium content of an aqueous stream where content of lithium and other species change with time according to another embodiment.

[0013] FIG. 3 is a flow diagram summarizing a method of recovering lithium from an aqueous source according to another embodiment.

[0014] FIG. 4 is a process flow diagram of a lithium recovery process that uses inertial density sensors to provide substantially continuous lithium concentration readings, according to one embodiment.

[0015] FIG. 5 is a flow diagram summarizing a method of extracting lithium from an aqueous lithium containing material, according to one embodiment.

[0016] FIG. 6 is a process flow diagram of a simulated moving bed system according to one embodiment.

[0017] FIG. 7 is a graph showing use of a proxy sensor in a system for extracting an element of interest from an aqueous material, according to one embodiment.

[0018] FIG. 8 is a process flow diagram of an extraction system according to another embodiment.

[0019] FIG. 9 is a process flow diagram of an extraction system according to another embodiment.

[0020] FIG. 10A is a graph showing data that can be obtained using sensors according to embodiments described herein. FIG. 10B is a close-up view of the graph of FIG. 10A.

[0021] FIG. 11 is a graph showing data like the data of FIG. 10, but with aberrations according to embodiments described herein.

[0022] FIG. 12A is a graph showing data similar to that of FIG. 11, but with more data aberrations. FIG. 12B is a close-up view of the graph of FIG. 12A.

DETAILED DESCRIPTION

[0023] It has been discovered that conductivity sensors can be used to monitor content of an element of interest in aqueous streams with sufficient accuracy and precision to allow a better management of the extraction methods. Sensors that reliably report concentration of other species than element of interest can be used to supplement conductivity measurements and improve accuracy of concentration reporting by such conductivity measuring devices.

[0024] Conductivity of an aqueous medium varies with ion content, so Total Dissolved Solids (TDS) can be ascertained from conductivity measurement as both parameters are correlated according to a linear relationship. Indication of the TDS provides an indication of the concentration of the element of interest. In an embodiment, contribution of other species to conductivity changes can be compensated by separately measuring content of those species and calibrating for coincident changes in concentration of the other species.

[0025] To measure TDS using such a device, solutions with known TDS can be prepared and provided to the device at different temperatures to obtain conductivity measurements. The readings of temperature and conductivity can be related to the known concentrations of lithium to yield a calibration relation. The calibration relation can be a linear equation based on regression, or other statistical treatment, or the calibration relation can be a table, in which conductivity and TDS can be interpolated.

[0026] Hereinabove, the example of a conductivity sensor to measure TDS has been provided as accurate and cost-efficient in-line water conductivity sensors exist. Moreover, such sensors may be installed without limitations on straight pipe diameters which enables to keep a compact installation and to obtain a great number of measurement points without increasing significantly the cost or complexity of a recovery system (including at least extraction, as well as optionally one or more of purification and concentration). However, other measurements may be used to infer the Total Dissolved Solids in a brine. For instance, density measurements such as using an inertial sensor or pressure sensor may be used and allow to obtain an indication of the TDS.

[0027] A difference between two aqueous materials can be used to infer a difference in TDS and hence, in the content of an element of interest, such as the lithium content. FIG. 1 is a flow diagram summarizing a method 100 according to one embodiment. The method 100 can be used to obtain a difference in TDS and concentration of an element of interest between two aqueous materials. At 102, a first parameter representative of the TDS of a first aqueous material containing an element of interest is obtained using a proxy sensor. The proxy sensor may include a conductivity sensor and the measured parameter may be conductivity. The proxy sensor is preferably an in-line conductivity sensor. The proxy sensor may alternatively include a density sensor (such as inertial density or pressure density)

[0028] At 104, a second parameter representative of the TDS of a second aqueous material is obtained using a proxy sensor. The same proxy sensor can be used, or a different conductivity sensor can be used. Where the same conductivity sensor is used to obtain the first and the second conductivity, appropriate flushing capability may be provided to avoid cross-contamination between samples of the two aqueous materials. Where two different sensors are used, the two

sensors can provide substantially continuous conductivity readings of the two aqueous materials. [0029] At **106**, the first parameter and the second parameter are compared to ascertain a difference between the conductivities or a difference in one or more parameters that depend on the conductivities (in particular, TDS). The difference can be used to monitor processing of the aqueous materials or to relate the compositions of the materials.

[0030] At **108**, optionally, the comparison may be used to infer a difference in concentration of an element of interest (such as lithium) between the first aqueous material and the second aqueous material. The method **100** can be for instance be useful where a process removes lithium from the first aqueous material to yield the second aqueous material. The comparison can provide a substantially continuous reading of the lithium removal.

[0031] Lithium concentration can be inferred from the parameter, and change in lithium concentration can be inferred from change in conductivity. If concentration of other species in the first and second aqueous materials that affect conductivity thereof are known, and temperature is known, calibration can be used to infer element of interest concentration of the first aqueous material from the first conductivity, and of the second aqueous material from the second conductivity. The two concentrations can be compared to understand the change in concentration of the element of interest between the two materials. Known solutions, with and without other species that affect conductivity, can be analyzed using the conductivity sensor or sensors to give calibration curves for inferring concentration of the element of interest from conductivity readings.

Additionally or alternately, additional independent sensors can be used to detect content of non-lithium species that affect the conductivity of the aqueous materials. Such additional sensors may be density sensors, pH sensors, spectrum sensors (e.g. IR or UV transmission spectrum analyzers), electrochemical sensors, optical (e.g. colorimetric) sensors, or a combination thereof.

[0032] FIG. **2** is a flow diagram summarizing a method **200** according to another embodiment. The method **200** can be used to determine content of an element of interest of an aqueous stream where content of the element of interest (hereinafter lithium for the sake of this example only) and other species change with time. At **202**, aqueous material containing lithium, and optionally other species, is provided to a conductivity sensor. At **204**, conductivity of the aqueous material is obtained from the conductivity sensor.

[0033] At **206**, an independent sensor, for example any of the sensors described above, is used to detect concentration of one or more other species in the aqueous material. Where changes in one non-lithium species predominate, or where content of all non-lithium species tend to vary together, or where only one non-lithium species is present, one sensor can be used to detect concentration of non-lithium species. Where changes in more than one non-lithium species need to be independently decoupled, more than one sensor may be used besides the conductivity sensor. Independent sensors that can be used include conductivity sensors, pH sensors, spectrum sensors (e.g. IR or UV transmission spectrum analyzers), electrochemical sensors, optical (e.g. colorimetric) sensors, or a combination thereof. Use of multiple, closely calibrated, and independent sensors, and averaging results can improve accuracy and reduce error. Results from multiple independent sensors detecting the same parameter can allow discovery of error conditions with a sensor so that sensor can be remediated.

[0034] At **208**, a multi-factor calibration relation is used to decouple lithium concentration from concentration of other species. Conductivity readings from the conductivity sensor are calibrated to multiple concentrations of lithium and non-lithium species, at different temperatures, to build the multi-factor calibration relation. When the multi-factor calibration relation is reduced to an equation relating conductivity with lithium concentration and concentration of other species, concentration of non-lithium species detected by a sensor other than conductivity can be substituted into the equation, and lithium concentration can be calculated from conductivity readings of the conductivity sensor.

[0035] FIG. **3** is a flow diagram summarizing a method **300** according to another embodiment. The

method **300** is a method of recovering lithium from an aqueous source.

[0036] At **302**, an aqueous lithium-bearing material is provided to an extraction stage for recovery of an element of interest from the aqueous material. In the following, the element of interest is lithium but the disclosure may be applied to any other element of interest, such as cobalt, nickel, copper or any other element of interest. The aqueous lithium-bearing material is obtained from a lithium source, which can be a surface source, such as a salar lake or a generated source (i.e. from washing lithium containing solid materials); a subterranean source, such as water produced from mines and wells; an industrial source, such as an aqueous lithium containing byproduct stream, an ocean source, or any other aqueous lithium source. The lithium source may be a solid or liquid material. For example, the aqueous lithium-bearing material may be obtained directly from a salar lake, or may be obtained by water washing of a lithium-bearing solid material.

[0037] The extraction stage uses any suitable method of extracting lithium from the aqueous lithium source to form an aqueous lithium extract. Lithium extraction is a stage in which the selective extraction of lithium is performed, i.e. the ratio of the lithium concentration and the TDS increases. The extraction stage can use any form of direct lithium extraction. For instance, in a direct lithium extraction process, an aqueous lithium-bearing material may be contacted with a lithium-selective medium to withdraw lithium ions from the aqueous lithium-bearing material into the lithium-selective medium, which may be a liquid or a solid. Such lithium-selective medium may be a sorbent or an ion exchange resin or a solvent. Another extraction process may include an electrochemical process.

[0038] A direct lithium extraction process may be an ion exchange or ion replacement process, where the withdrawal medium is pre-loaded with ions that are exchanged to the feed fluid while withdrawing other ions from the feed fluid. In some cases, the direct lithium extraction process may be an adsorption process where ions are adsorbed from the aqueous lithium stream solution onto the surface of a solid adsorbent material that is selective to lithium, such as metal oxide, metal hydroxide or such material mixed with a resin. In other cases, the direct lithium extraction process may be an absorption process where ions are absorbed from the brine solution into the bulk of a solid absorbent material that is selective to lithium. A desorbent solution is used in all of these cases, as well. These cases of pure sorption-desorption can require regeneration of the withdrawal medium because unloading of ions from medium is not quantitative.

[0039] Solid lithium selective media, such as a resin treated, coated, or impregnated with materials such as aluminum hydroxide, manganese oxide, or titanium oxide can be used. Phosphorus-based liquid lithium-selective media, such as the LiSX™ solvent extraction medium available from Tenova SpA of Castellanza, Italy, or the CYANEX® 936P extractant available from Solvay S.A. of Brussels, Belgium, can also be used. Other effective liquid extractants have also been reported, such as 3-benzoyl-1,1,1-trifluoroacetone dissolved in 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide.

[0040] In a sorption/desorption embodiment where the adsorbent material is a solid, the adsorbent material may be stationary or fluidized within the vessel, or conveyed through one or more vessels or zones for contacting with the brine, for example in a counter-current format. In particular, the adsorbent material may be contained in a plurality of vessels in flow communication with one another and the vessels may be fluidly connected with a plurality of zones (ie inlets/outlets) during the extraction process. The extraction **104** may therefore take place continuously, for instance loading resin in a first vessel with lithium by fluidly connecting this vessel with the brine source while unloading resin in a second vessel by fluidly connecting the second vessel with the eluent and washing a third vessel using a strip solution. The extraction may be continuous counter-current adsorption desorption (CCAD). An example of a counter-current adsorption desorption that may be used is for instance described in U.S. Pat. No. 11,365,128 from EnergySource Minerals, which describes an example process.

[0041] When the lithium-selective medium is loaded with lithium, an eluent is used to remove the

lithium from the lithium-selective medium to form a lithium extract. Concentration of lithium in the lithium extract can be varied and controlled by adjusting a flow rate of the eluent used to remove lithium from the lithium-selective medium. The flow rate can be adjusted to achieve a target lithium concentration in the lithium extract. In the liquid medium context, mixing of any convenient sort can be applied during loading and unloading of the lithium-selective medium. [0042] Direct lithium extraction processes can also use a lithium selective electrochemical separation process. A lithium selective electrochemical separation process uses a voltage bias to drive materials through a lithium selective membrane to separate lithium from an aqueous lithium source. In an exemplary embodiment, the aqueous lithium source is brought into contact with a first side of the lithium selective membrane, and an aqueous eluent material is brought into contact with a second side of the lithium selective membrane, opposite from the first side. The voltage bias is applied within the aqueous lithium source and the aqueous eluent material to form an electric field within both materials and extending across the lithium selective membrane. The electric field provides a driving force to move, or increase movement of, charged species through the lithium selective membrane. The species motivated by the electric field to move through the lithium selective membrane depends on the configuration of the lithium selective membrane. For example, the lithium selective membrane may selectively pass lithium ions more than other ions or the lithium selective membrane may selectively block passage of lithium ions more than other ions. [0043] Direct lithium extraction processes that include lithium selective electrochemical separation processes use lithium selective membranes. Such membranes can include, or be made of, lithium selective materials such as lithium aluminum germanium phosphate, lithium aluminum titanium phosphate, lithium lanthanum titanates, or a metal organic framework type material such as UiO-66 with acid and amine groups. Such materials can be configured alone in a membrane structure or can be added to a support material, such as a resin, configured into a membrane structure. [0044] In case of such direct lithium extraction processes, the flow rates of the aqueous material and/or eluent as well as the applied voltage, for instance, may be tuned to obtain a more efficient lithium extraction.

[0045] At **304**, the extract is provided to a processing stage to transform the extract into a product incorporating the extracted metal. In a lithium context, the product can be a concentrate, for example a concentrated lithium chloride solution or slurry or a lithium carbonate solution or slurry, for further processing and/or purification, or the product may be a battery raw material such as a lithium hydroxide solution, slurry, or solid. The processing stage can use one or more of a concentration process (i.e. increasing the total dissolved solids), a purification process (i.e. decreasing the ratio of impurity to element of interest), or a conversion process (converting a first product including the element of interest to a second product including the element of interest, for instance lithium chloride to lithium hydroxide). A concentration process can involve evaporation, for example flashing or other vaporization processes, membrane separation, including reverse osmosis, osmotically assisted reverse osmosis, nanofiltration, membrane distillation, etc., to remove water from the extract to form a concentrate. The purification process may include precipitation, filtration, chemical reaction, impurity ion withdrawal, or a combination thereof. For instance, as an example, it can include coagulation-flocculation and filtration and then ion exchange for remove divalent ions. The conversion process typically uses a reagent to react with the target ions in the extract, or another stream such as a concentrate stream, to form the lithium product.

[0046] In one embodiment, an extract is provided to a concentrator that removes water from the extract by membrane separation, for instance reverse osmosis and/or counter-flow reverse osmosis and/or osmotically assisted reverse osmosis. The concentrate is provided to a conversion process that converts the lithium in the concentrate to lithium hydroxide, either via a lithium carbonate intermediate, or directly to lithium hydroxide in an electrochemical process.

[0047] The processes of the processing stage may form solids. For example, concentration may

result in lithium salts precipitating if the solubility limit of the lithium salts are reached. In another example, a conversion may add ions that preferentially precipitate one or more lithium salts having lower solubility, forming solids. In any solids-forming process, the processing stage can include solids handling and removal processes, such as filtration or cyclonic processes, to capture and control solids that may be formed in the processing stage. Other processing stages that can be used, instead of or in addition to, the examples described above include ion exchange units, adsorption/desorption units, filtration units, membrane separation units, and moving bed or simulated moving bed units.

[0048] At **306**, a sensor is used to determine a parameter linked to one or more materials of the extraction stage and/or the processing stage and infer TDS based on such parameter. Such sensor may be a conductivity sensor as has been discussed hereinabove. It may also be a density sensor such as an inertial density sensor and/or a pressure sensor. The material may be any or all of the aqueous lithium source, the lithium extract, or any other stream of any embodiment of the process, such as the eluent, the lithium depleted stream, the lithium concentrate stream, the lithium product stream, and any byproduct streams. In one example, a water stream is obtained from the processing stage and routed to the extraction stage for use as an eluent or a diluent. A sensor can be used to determine TDS of the water stream and one or more parameters of the processing stage can be adjusted based on the TDS of the water stream. In some embodiment, the lithium concentration may be inferred from the TDS and possibly other measurements as explained hereinabove. TDS of other streams of the extraction and processing stages may also be determined using the same or other similar sensors and the readings may be compared to infer changes in TDS, and in some cases in lithium concentration at different locations. The extraction stage and/or the processing stage can be operated based on the readings from the sensors.

[0049] In the method **300**, an additional sensor can be used, as described above to sense concentration of one or more non-lithium species to provide, or improve, lithium concentration detection. The sensors may be collocated, and signals from both sensors routed to a controller to decouple the concentrations of lithium and non-lithium species, which can both be reported and separately controlled.

[0050] The methods **100** and **200** can be used with the method **300** to operate the lithium recovery process. In the following examples that relate to a direct lithium extraction using a selective lithium media are provided. In one case, a sensor is used to determine TDS via a proxy measurement such as conductivity, or lithium concentration, of a lithium-depleted stream of a direct lithium extraction process resulting for instance from withdrawal of lithium ions from an aqueous lithium source into a solid lithium-selective medium. TDS in the lithium-depleted medium can be monitored, and rise in TDS that cannot be attributed to a change in flow rate or to lithium concentration of the aqueous lithium-bearing material, can be understood as increasing lithium loading of the ion withdrawal medium resulting in incipient breakthrough of lithium to the lithium-depleted stream. Depending on the lithium concentration in the lithium-depleted material, contacting the aqueous lithium-bearing material with the lithium-selective medium can be discontinued if the lithium concentration indicates an endpoint has been reached. The direct lithium extraction process can be monitored and operated based on the substantially continuous readings of TDS or of the proxy measurement such as conductivity, and differences in the readings.

[0051] When the lithium-selective medium is loaded with lithium, as indicated by lithium beginning to break through the lithium-selective medium to the lithium-depleted stream, flow of the aqueous lithium-bearing material to the lithium-selective medium can be discontinued, and an eluent can be contacted with the lithium-selective medium to remove lithium from the lithium-selective medium, thus forming the lithium extract.

[0052] A similar process of determining lithium concentration of the lithium extract can be used to determine when the lithium-selective medium has been unloaded to the extent that removal of lithium from the medium is slowing. TDS and/or lithium concentration of the lithium extract and

the eluent can be determined as described herein, using a proxy sensor such as a conductivity or density sensor, and optionally an additional sensor for non-lithium species. When lithium concentration in the eluent compared with lithium concentration in the lithium extract indicates lithium removal from the lithium-selective medium is slowing, flow of the eluent to the lithium-selective medium can be discontinued or reduced, and flow of the aqueous lithium-bearing material restarted for a new cycle. The proxy sensors can be repeatedly used in this way to detect loading and unloading endpoints.

[0053] The various endpoints described above can be defined by comparison to standards. For example, TDS and/or lithium concentration in the lithium-depleted stream can be compared to a standard to determine whether an endpoint has been reached. Alternately, or additionally, difference in the proxy measurement such as conductivity or density can be compared to a standard to identify the endpoint. In one method, when TDS and/or lithium concentration in the lithium-depleted stream rises to or above a standard value, or approaches the standard, flow of the aqueous lithium-bearing material to the lithium-selective medium can be discontinued and flow of the eluent started. Likewise, when TDS and/or lithium concentration in the lithium extract falls to or below a standard value, or approaches the standard, where the standard for the lithium-depleted stream is a first standard and the standard for the lithium extract is a second standard, flow of the eluent to the lithium-selective medium can be discontinued and flow of the aqueous lithium-bearing material can be restarted. In another method, when absolute value of a difference in the proxy measurement (such as conductivity or density) between the aqueous lithium-bearing material and the lithium-depleted material falls to or below a difference standard, or approaches the difference standard, flow of the aqueous lithium-bearing material to the lithium-selective medium can be discontinued and flow of the eluent started. Likewise, when absolute value of a difference between the eluent and the lithium extract falls to or below a difference standard, or approaches the difference standard, flow of the eluent can be discontinued, and flow of the aqueous lithium-bearing material restarted. In this case, the same difference standard can be used during the loading and unloading phases, or different difference standards can be used for the two phases.

[0054] The substantially continuous monitoring described herein provides the ability to monitor loading capacity of the lithium-selective medium. If a proxy sensor is used to monitor TDS of the aqueous lithium-bearing material and the lithium-depleted aqueous material, the difference between the materials can indicate that loading capacity of the lithium-selective medium is falling as the medium collects lithium ions. If lithium concentration is determined, for example using additional sensors to report concentration of non-lithium species, and if flow rate of one or both streams is known, then an actual mass-loading of lithium within the lithium-selective medium can be calculated. The mass-loading of each cycle can be archived for analysis, and a decline in mass-loading per cycle, or mass-loading per cycle reaching or approaching a minimum standard, can indicate that regeneration of the lithium-selective medium is needed. Same approach may be applied to mass-unloading by monitoring the lithium extract and the eluent. Regeneration of some lithium-selective media can be performed by exposure to hot water and/or hot gas, which can be done by flowing through the lithium-selective medium, soaking the lithium-selective medium in quiescent hot water, or both. Other media can be regenerated by exposure to reagents that remove lithium. Alternately, or additionally, loading and unloading cycle time can be monitored for indications that regeneration is needed. Decreasing cycle time, or cycle time reaching or approaching a minimum standard, can indicate that a regeneration cycle should be started.

[0055] In some cases, a direct lithium extraction process uses a plurality of extractors, each having a lithium-selective medium for contacting with an aqueous lithium-bearing material. Proxy sensors can be used to monitor operation of each extractor to determine when each extractor reaches a loading endpoint and an unloading endpoint by substantially continuous reporting of TDS-related parameter and/or lithium concentration of the input and output streams of each extractor.

[0056] Substantially continuous monitoring of density and/or lithium concentration in streams of

the extraction stage can also be used to optimize extraction performance. Any or all of temperature, pressure, flow rate, pH, cycle time, residence time, moving speed of the lithium-selective medium in a moving bed or simulated moving bed application, or other parameters may be adjusted depending on the proxy measurement, change in the proxy measurement over time, difference in the proxy measurement taken between two or more materials, lithium concentration, change in lithium concentration, rate of change of lithium concentration, distance from an endpoint, cycle time, mass-loading of the lithium-selective medium, or trend thereof.

[0057] As noted above, proxy sensors can be used for substantially continuous monitoring of TDS and/or lithium concentration via proxy measurements (such as conductivity or density) in any stream of a lithium recovery process. For example, in a concentration process where lithium concentration is increased in an aqueous material to yield a lithium concentrate, TDS and/or lithium concentration in a lithium concentrate stream can be monitored and operation of the concentration process adjusted to optimize or improve performance of the concentration process. Thus, temperature, pressure, flow rate, residence time, or other parameters can be adjusted to achieve a target composition of the lithium concentrate. Likewise, proxy sensors can be used to monitor performance of a lithium conversion process. Because the lithium conversion process involves changing compositions of many species, additional sensors can be used to monitor multiple non-lithium species, while inertial density sensors are used to monitor density and/or lithium concentration. Using such sensors, conversion of lithium in a feed to the conversion process can be monitored substantially continuously and operation of the conversion process can be adjusted to optimize or improve results.

[0058] FIG. 4 is a process flow diagram of a lithium recovery process **400**, according to one embodiment. The lithium recovery process **400** uses proxy sensors to provide substantially continuous TDS and/or lithium concentration readings based on proxy measurements for multiple lithium-bearing streams, and a controller adjusts the process **400** based on the readings, among other signals.

[0059] The lithium recovery process **400** has an extraction stage **402** and a processing stage **404**. The extraction stage extracts lithium from an aqueous lithium-bearing material **406** to yield a lithium extract **408** and a lithium-depleted material **410**. In this case, the extraction stage uses direct extraction with a solid lithium-selective ion withdrawal medium, so an aqueous eluent **412** is provided to the extraction stage **402** to unload lithium from the medium. The lithium-selective medium withdraws lithium from the aqueous lithium-bearing material to form the lithium-depleted material and the eluent desorbs lithium from the loaded lithium-selective medium to yield a lithium extract **408**.

[0060] A first proxy sensor **414** is coupled to the lithium-depleted material **410**, and a second proxy sensor **416** is coupled to the lithium extract **408**, to provide substantially continuous readout of density and/or lithium concentration so that loading and unloading of the lithium-selective medium can be tracked. A controller **418** is operatively coupled to the first and second proxy sensors **414** and **416** and to the extraction stage **402** to control operation of the extraction stage **402**, for example loading and unloading start and stop and operating parameters like temperature, pressure, flow rate, and moving speed in a moving bed or simulated moving bed application, based on signals from the inertial density sensors **414** and **416**. As discussed above, the proxy sensor may be a conductivity sensor and/or a density sensor such as an inertial sensor or a pressure sensor that provides a parameter based on which the TDS and optionally the lithium concentration may be inferred.

[0061] The controller **418** can be configured to interpret signals from the proxy sensors **414** and **416** to resolve TDS and/or lithium concentration in the lithium-depleted material **410** and the lithium extract **408**. Alternately or additionally, the controller **418** and/or proxy sensors **414** and **416** may be configured with computing capability to calculate TDS and/or lithium concentration of the lithium-depleted material **410** and the lithium extract **408**, using a calibration relation and/or

using signals from additional sensors, as described above.

[0062] An optional third proxy sensor **420** can be coupled to the aqueous lithium-bearing material **406**, and operatively coupled to the controller **418**, to provide substantially continuous readout of TDS and/or lithium concentration in the aqueous lithium-bearing material **406**, so that composition in the various streams can be compared and balanced to understand performance of the extraction stage **402**. For example, as noted above, the controller **418** can be configured to compute and track parameters such as mass-loading of the lithium-selective medium, loading and unloading cycle time, and comparison to standards.

[0063] The processing stage **404** transforms the lithium extract **408** into a lithium product **421** and a byproduct **422**. Depending on the operations performed in the processing stage **404**, the lithium product **421** may be a lithium chloride product, a lithium carbonate product, a lithium hydroxide product, or a mixture. In one case, the processing stage **404** has a concentrator that removes water from the lithium extract **408** to yield a lithium concentrate and a water stream. The byproduct **422** may be, or may include the water stream. An optional fourth proxy sensor **424** is coupled to the lithium product **421**, and operatively coupled to the controller **418**, to provide substantially continuous readings of TDS and/or lithium concentration in the lithium product **421**. The fourth proxy sensor **424** is configured and calibrated based on the composition of the lithium product **421**, whether a lithium chloride product, a lithium carbonate product, a lithium hydroxide product, or a mixture thereof. The processing stage can also include a conversion unit to convert the lithium in the lithium extract **408**, the lithium concentrate produced by the concentrator, or both in any mixture, into the lithium product **421**. For example, the conversion unit may use sodium carbonate to convert lithium chloride to lithium carbonate. In such a case, the byproduct **422** will include sodium, and can be characterized as primarily a sodium chloride stream. In another example, lithium carbonate can be converted to lithium hydroxide by reaction with calcium oxide or calcium hydroxide. In such a case, the byproduct **422** will include calcium, which may precipitate as calcium carbonate.

[0064] Optionally, a fifth proxy sensor **426** can be coupled to the eluent **412**, and operatively coupled to the controller **418**, to determine TDS and/or lithium concentration of the eluent **412**. As described above, TDS of the eluent can be compared with TDS of the lithium extract **408** to monitor unloading of lithium ions from the lithium-selective medium of the extraction stage **402** and to determine when the unit can be switched from unloading to loading mode by discontinuing flow of the eluent and restarting flow of the aqueous lithium-bearing material **406**. Parameters related to performance of the lithium-selective medium like extent of mass-loading and-unloading can also be determined and tracked to determine when the lithium-selective medium needs to be regenerated, for example when a degree or amount of mass-loading before the lithium-selective medium stops withdrawing lithium ions from the aqueous lithium-bearing material falls to near, at, or below a standard, or when a degree or amount of mass-unloading before the lithium-selective medium stops releasing lithium ions to the eluent falls to near, at, or below a standard.

[0065] Where appropriate and useful, some or all of the byproduct **422** can be routed to the extraction stage **402** for use as, or with, the eluent **412**. An optional sixth proxy sensor **430** can be coupled to the recycled portion of the byproduct **422**, and operatively coupled to the controller **418**, to determine TDS and/or lithium concentration of the recycled portion of the byproduct **422** to control composition of the eluent **412**. In some cases, a small concentration of lithium ions in the eluent **412** can be helpful to performance of the extraction stage **402**. The controller **418** can be configured to adjust flow rates of the eluent **412** and the recycled portion of the byproduct **422**, based on TDS and/or lithium concentration determined by the fifth proxy sensor **426**, the sixth proxy sensor **430**, or both, to target a composition of the eluent **412** for best results in the extraction stage **402**.

[0066] The process **400** uses proxy sensors to provide substantially continuous readings of a proxy measurement such as conductivity or density and infer TDS and/or lithium concentration in one or

many materials of the process **400**. Materials other than the materials indicated in FIG. **4** can be monitored using proxy sensors. For example, multiple feed streams and eluent streams of the extraction stage **402** can be monitored if the extraction stage contains multiple extraction units. Multiple materials of the processing stage **404**, such as lithium concentrate streams, converted lithium streams, and streams separated in the processing stage **404** such as byproduct streams, can also be monitored using proxy sensors. The controller **418** can be configured and operatively coupled to all the proxy sensors to control multiple aspects of the operation of the process **400** to achieve desired results in the lithium product **421**, or desired operating profiles of the process **400**, such as energy efficiency and environmental impact. As an example of the latter, the controller **418** can be configured to adjust flow rate of the lithium-depleted material **410** to the environment or to remediation before being returned to the environment, based on density and/or lithium concentration determined by the first inertial density sensor **414**.

[0067] Where necessary, an additional sensor **428** can be co-located with, or coupled to the same material as, each inertial density sensor of the process **400**, to provide independent detection of other species in each material that may affect changes to the density of the material, so those effects can be decoupled. The controller **418** can be configured to, and operatively coupled with, the additional sensors **428** to receive signals representing species to be detected by each additional sensor **428** and to decouple concentration of such species from lithium concentration using a multi-factor calibration relation, as described above.

[0068] It should be noted that, where a single controller is operatively coupled to more than one proxy sensor, accuracy of comparing readings from the plurality of proxy sensors can be improved. For example, the same calibrant fluid can be provided to more than one proxy sensor, operatively coupled to a single controller, at the same time, and the controller can be calibrated to remove any bias between the readings from the inertial density sensors. In this way, accuracy of differential TDS measurements using proxy sensors can be improved.

[0069] In an embodiment shown on FIG. **5**, the method includes continuously extracting **502** an element of interest using a cyclical simulated moving bed process. In such a process, as shown on FIG. **6** that describes a simple simulated moving beds system **600**, vessels **602** including the selective medium are fluidly connected, in a cyclical pattern, to different stations S.sub.1, . . . S.sub.n (here 10) distributed in several zones to perform different operations on the media in the vessels. The stations represent operations that are performed on the contents of each vessel. In an embodiment, the zones include a loading zone **604**, a brine displacement zone **606**, a stripping zone **608** and a strip displacement zone **610**. In the loading zone, the brine (stored in a brine tank **612**) circulates through the vessels to load the element of interest within the selective medium contained in the vessels. Fluid connections are established to route the brine through all the vessels fluidly connected sequentially within the brine zone operation. In the brine displacement zone **606** a fluid is used to displace the brine from the vessels and to push back the brine in tank **612**. The system **600** is configured such that two vessels are operated in the brine displacement zone **606** such that the fluid used for displacing the brine flows sequentially through two vessels, one in station (or mode) **S1** and another in **S2**. In the stripping zone **608**, an eluent (stored in a tank **614**) circulates through the vessels to unload the element of interest from the selective medium contained in the vessel and store it in a product tank **616**. The system **600** is configured such that three vessels are operated in the stripping zone **608**. In the strip displacement zone **610**, the lithium depleted stream **618** exiting from the loading zone **604** is used to displace the eluent from the vessels operating in the strip displacement zone **620** so that the eluent is returned to the eluent tank **614**. These various processes are accomplished through fluid connections that are operated to change the lineup of the vessels according to the cyclical pattern. Each vessel is fluidly connected according to each operating station in sequence, cyclically, so each vessel is processed through each position of each zone cyclically. Thus, the fluid connections are operated such that the vessel at station (or mode) **S1** is connected to receive fluid from another vessel at station (or mode) **S1** and to return fluid to the

tank **612**, as station (or mode) **S2**. All vessels of the system **600** are cycled through the different operating modes, or stations, sequentially in this manner. Hence, the fluid circulates cyclically through the different zones, in the following pattern: the loading zone then the strip displacement zone, then the strip zone, then the brine displacement zone, i.e. from a first station **S.sub.1** to a last station **S.sub.n**, then returning again to operate at the first station **S.sub.1**.

[0070] Each vessel takes the different stations **S.sub.1**, . . . **S.sub.n** of each zone, by manipulating fluid connections to the vessels, so that the change of operation of each vessel, and the change of treatment of the contents of each vessel, is counter-current to the flow of the fluids (i.e. the vessel receives fluid circulation in the loading zone, then the strip displacement zone, then the stripping zone and then the brine displacement zone, i.e. in a pattern from operating stations **S.sub.n** to **S.sub.1**). Each zone may include several operating stations in series as shown on FIG. **6** and/or in parallel. Any number of stations and/or zones, and stations per zone, can be used. Note that in most cases the vessels do not move but that the operation performed on the contents of each vessel relative to the eluent or brine tank changes due to the change of the fluidic connections between the vessels and said tanks. A fluid circulation device (not shown) such as a pump is able to control the flow rate through each of the zones, and valves such as multi-port valves can be used to control the fluidic connections. The stations are thus analogous to operating modes of the vessels. Each operating mode is different by virtue of the fluids flowing through a vessel and by the vessel's order in the fluid flow.

[0071] For example, stations **S3-S6** are sequential fluid flow stations (or modes) operated in the loading zone **604**, where brine flows from the brine tank **612** through the vessels of the stations **S3-S6** sequentially to load the media of the vessels in those stations with material from the brine. After operating vessels in stations **S3-S6** for a duration, optionally until an end point is recognized, fluid connections of the vessels are changed such that the vessel at the **S6** station, the last station to receive fluid from the tank **612**, is fluidly connected to provide fluid to another vessel connected at the **S6** station (or mode) and is now in station (or mode) **S5**, the third station to receive the fluid originating from the brine tank **612**. The vessel now in station **S5** was partially loaded with elements of interest in station **S6** and continues absorbing elements of interest in station **S5**. As the vessel is cycled into stations **S4** and **S3**, loading of the material in the vessel increases, reaching a maximum during processing in the stations (or modes) **S3-S6**. Also, as the vessel is cycled into stations **S4** and **S3**, the concentration of the elements of interest in the brine increases compared to when the vessel is in station **S5** or **S6** as, in **S5** or **S6**, the elements are partially absorbed by the selective media of the vessels at earlier loading stations **S3** and **S4**.

[0072] The method includes using proxy sensor **504** to obtain a first value of a proxy parameter (such as conductivity) representative of the TDS of a material used in the process at a first time. The proxy sensors may be placed at the entrance and/or the exit of each vessel (or of a group of vessels, at minimum of one vessel). Optionally, the proxy sensors are disposed in the vessels so that there is one proxy sensor in each of the vessels. In an embodiment, the proxy sensor is a conductivity measurement sensor that may be easily disposed in each of the vessels. It has to be noted that the proxy sensors may also be disposed in the fluidic connections fluidly connecting the vessels. A controller may be connected to each of the proxy sensor and is able to associate each sensor with a station using other measurements provided by additional sensors (such as sensing positions of the fluidic connections). Other additional measurements to determine other parameters of the system (such as flow rate) may be obtained by the system.

[0073] The method also includes using a proxy sensor **504**—the same or another—to obtain a second value of the proxy parameter (such as conductivity) representative of the TDS of the material used in the process at a second time. Optionally proxy measurements are taken in other locations (as can be see on FIG. **7**). However, obtaining the proxy measurements in all locations over time is not needed. It may be sufficient to obtain the value of the conductivity at target locations of the process over time.

[0074] An example of a proxy measurement is provided in FIG. 7. In this example, the system shows the conductivity versus time taken in a single vessel within a full cycle. Such measurement may be related to the vessel position in the cycle as this information is available from other sensors. For this graph, the vessel has spent the same time in each of the stations. We can see that during the loading phase **604** the conductivity (ie TDS) is high while during the brine displacement **606** conductivity decreases as only the loaded elements remain in the vessel. During strip phase **608**, conductivity decreases too as the loaded elements of interest are stripped from the selective medium and during the strip displacement **610** phase the conductivity and TDS remain approximately constant.

[0075] The method according to the disclosure includes adjusting **508** one or more parameters of the cyclical simulated moving bed process based on the first and second values of the measured parameter. The one or more parameters may for instance include the flow rates of each of the zones of the simulated moving bed process. It is not necessary to obtain the proxy parameter for the full cycle.

[0076] In an embodiment, the method includes first adjusting the strip displacement rate. The goal of strip displacement is to displace as much eluent (low conductivity) into the eluent tank using the lithium depleted stream (high conductivity) without getting any brine into the strip tank. The ideal strip displacement rate is obtained when the highest volume displacement is achieved before seeing a high conductivity on the outlet of the first station of the strip displacement zone, i.e. when the lowest conductivity is measured at the outlet corresponding to station **S6** in FIG. 6.

[0077] A strip displacement rate may be incrementally increased until the conductivity increases. Optionally strip displacement rate may then be adjusted back in small increments until only low conductivity is detected at said point.

[0078] Optionally, the parameter representative of the TDS may be measured for each column during a full cycle and adjusting the strip displacement rate may be performed for each position of the vessels. This enables us to maximize the efficiency of the process. Indeed, the optimal strip displacement rate may vary depending on the specific void volume of each vessel, which can be affected by the amount of resin in the column, differences in connecting pipe lengths, specific permeability through each resin bed.

[0079] In an embodiment, the void volume for each column is calculated using the above-mentioned process, enabling to derive the total void volume for each zone in each position of the cycle. The brine displacement flow rate may then be set in each position taking the corresponding void volume of the zone as a reference: for instance, by choosing a flow rate that enables a predetermined percentage of the void volume while the vessels are in a given position. In an embodiment, the flow rate in the brine displacement zone is adjusted to maximize the conductivity at the outlet of the brine displacement zone (outlet of station **S2** in FIG. 6)

[0080] In an embodiment, the stripping flow rate is determined based on desired final concentration and known adsorption capacity of the resin. The flow rate is adjusted to maximize the conductivity at the outlet of the strip zone (outlet of the station **S10** in FIG. 6).

[0081] In an embodiment, the loading flow rate can be tuned so that the resin is fully loaded with the element of interest and the lithium depleted stream **618** (i.e. at the outlet of station **S5** in the system **600**) does not have any element of interest in it. In this case, the flow rate is optimized so as to minimize the flow for which there is no conductivity drop. The flow rate may be first dropped down until a conductivity drop is seen and then incrementally increased until such conductivity drop does not occur.

[0082] Similarly as what has been discussed for the strip displacement zone, the flow rates for each zone may be optimized for each position of the system, i.e. for a first position of the system, i.e. if a vessel **602.sub.1** is in station **S1** and vessel **602.sub.n** is in station **Sn**, the flow rate of each of the zone is to an initial predetermined value. If the system takes a second position, with vessel **602.sub.1** is in station **Sn** and vessel **602.sub.n** in station **S.sub.n-1**, the flow rates of each of the

zones are set to a second value, that may be equal or different from the first value. The flow rates of each of the zones are of course determined independently from each other.

[0083] Monitoring the parameters through time at each cycle enables to detect default in the simulated moving bed system, such as leak, for instance by comparing the flow rates in relationship with one particular vessel in one particular station at several different times.

[0084] As explained above, the monitoring of the proxy parameter (such as conductivity) is not limited to extraction stage, or simulated moving bed process.

[0085] Extraction of elements of interest may include additional stages as explained above such as concentration and/or purification. In the case of purification, the proxy parameter measurement helps to determine the efficiency of the purification stage as the TDS drop in the feed containing the element of interest and the TDS increase in another feed, such as an eluent feed are representative of such efficiency. Parameters, such as flow rate, may also be varied to maximize efficiency of such stages.

[0086] Concentration stages may also be optimized using proxy measurements. In this case the total TDS of the feed containing the element of interest is increased resulting in higher value of the proxy parameter. In this case, the parameters may be varied so that the ratio between the value of the proxy parameter for the feed exiting the concentration stage and the value of the proxy parameter for the feed entering the concentration stage is maximized.

[0087] The methods described herein can be applied to any system having multiple vessels to extract an element of interest, such as lithium, from an aqueous material. FIG. 8 is a process flow diagram of a system **800**, according to another embodiment. The system **800** is a system for extracting an element of interest from an aqueous material, like the other systems described herein. The system **800** uses multiple vessels operated in a cyclic, permuted pattern, to extract the element of interest from the aqueous material. The system **800** has a first vessel **802A** and a second vessel **802B**, each containing a selective medium for extracting the element of interest from the aqueous medium. An aqueous medium source **804** provides the aqueous medium, and is fluidly coupled to the vessels **802A** and **802B** by a flow system **806**. The flow system **806** also fluidly couples an eluent source **808** to the vessels **802A** and **802B**. The flow system **806** contains piping and valves that enable routing the aqueous medium from the aqueous medium source **804** to the first vessel **802A** or to the second vessel **802B**. The flow system **806** also contains piping and valves that enable routing the eluent from the eluent source **808** to the first vessel **802A** or to the second vessel **802B**. Thus, each of the aqueous medium source **804** and the eluent source **808**, as a target fluid, is fluidly couplable to the first vessel **802A** and/or the second vessel **802B**, individually or together, concurrently or separately. A first pump **810A** drives flow of the aqueous medium from the aqueous medium source **804** to the vessels **802A** and **802B** and a second pump **810B** drives flow of the eluent from the eluent source **808** to the vessels **802A** and **802B**.

[0088] A controller **850** is operatively coupled to the flow system **806** to control the flow system **806**. The controller **850** is configured to control the flow system **806**, for example by manipulating the valves of the flow system **806**, to route the aqueous material from the aqueous material source **804** to the first vessel **802A** while routing the eluent from the eluent source **808** to the second vessel **802B**, and to route the aqueous material from the aqueous material source **804** to the second vessel **802B** while routing the eluent from the eluent source **808** to the first vessel **802A**. By operation of the controller **850**, so configured, the first vessel **802A** can absorb the element of interest from the aqueous medium while the second vessel **802B** is releasing the element of interest into the eluent, and vice versa. The controller **850** is configured to control the flow system **806** to independently flow and route the aqueous material source **804** and the eluent source **808**. For example, the controller **850** can be configured to stop flow from the aqueous material source **804** altogether while allowing flow from the eluent source **808** to the first vessel **802A**, the second vessel **802B**, or both. The controller **850** can also be configured to stop flow from the eluent source **808** altogether while allowing flow from the aqueous material source **804** to the first vessel **802A**,

the second vessel **802B**, or both.

[0089] Thus the first pump **810A** can be configured to pump the aqueous medium to the first vessel **802A** or the second vessel **802B** and the second pump **810B** can be configured to pump the eluent to the first vessel **802A** or the second vessel **802B**, and the controller **850** can be further configured to control the flow system **806** by switching the effluent of the first pump **810A** between the first vessel **802A** and the second vessel **802B** and by switching the effluent of the second pump **810B** between the first vessel **802A** and the second vessel **802B**.

[0090] The system **800** can have a first sensor **812A** and a second sensor **812B**, each sensor to measure a parameter of one or more streams at the first and second vessels **802A** and **802B**. The first sensor **812A** is operatively coupled to the first vessel **802A**, or to one or more streams at an inlet or outlet of the first vessel **802A**. The second sensor **812B** is operatively coupled to the second vessel **802B**, or to one or more streams at an inlet or outlet of the second vessel **802B**. The sensors **812A** and **812B** measure characteristics of the materials flowing into and/or out of the first and second vessels **802A** and **802B**. The controller **850** can be operatively coupled to the first and second sensors **812A** and **812B** to receive signals, for example a first signal from the first sensor **812A** and a second signal from the second sensor **812B**, that represent a characteristic of the material sensed by the sensors. The characteristics can be total dissolved solids, or a parameter derived from, or representing, total dissolved solids. Each of the first sensor **812A** and the second sensor **812B** can be a single sensor or a sensor station comprising a plurality of independent or interdependent sensors. Each of the first sensor **812A** and the second sensor **812B** can be a continuous, in-line sensor, or plurality of sensors, that measures one or more of temperature, pressure, density, or conductivity. Instruments such as thermocouples, diaphragms, inertial flow sensors such as Coriolis sensors, nuclear absorption sensors, and electrical current sensors can be used. The first sensor **812A** is shown as a single sensor, or sensor station, to measure one or more characteristics of the aqueous medium and the eluent at the first vessel **802A**, but the first sensor **812A** can be multiple independent or interdependent sensors, each individually operatively coupled to the aqueous material or the eluent. Likewise, while the second sensor **812B** is shown as a single sensor for measuring the aqueous medium and the eluent at the second vessel **802B**, the second sensor **812B** can be multiple independent or interdependent sensors, each individually operatively coupled to the aqueous material or the eluent.

[0091] The controller **850** can be configured to control operation of the system **850** based on signals received from the sensors **812**. In one case, the controller **850** is operatively coupled to the valves of the flow system **806** to control a flow rate of the first vessel **802A** based on a signal received from the first sensor **812A** and to control a flow rate of the second vessel **802B** based on a signal received from the second sensor **812B**. The signals received from the sensors **812** can indicate performance of the selective medium within each vessel in removing the element of interest from the aqueous medium and in releasing the element of interest into the eluent. Where the signals from the sensors **812** indicate such performance does not meet a standard, flow rate of each vessel can be adjusted to bring the measured characteristic to the standard. In one case, the sensors **812** can be located at outlets of the vessels **802A** and **802B** to measure characteristics of the effluent of the vessels. Thus, one sensor **812** can be located at an effluent of the first vessel **802A**, or the second vessel **802B**, where a depleted stream, after removal of the element of interest from the aqueous medium, exits the vessel **802A** or **802B**. The measured characteristic of the depleted stream can be related to effectiveness of the selective medium within the respective vessel **802** in removing the element of interest from the aqueous medium. Likewise, a sensor **812** can be located at an effluent of the first vessel **802A**, or the second vessel **802B**, where a product stream, after the element of interest is released from the selective medium into the eluent, exits the respective vessel. The measured characteristic of the product stream can be related to effectiveness of the selective medium within the respective vessel **802** in extracting the element of interest, for example effectiveness in releasing the element of interest. The controller **850** can be configured to adjust

flow rates at the respective vessels based on the respective signals to achieve a target effectiveness of the process performed by the system **800**.

[0092] The vessels **802A** and **802B** are configured to receive the aqueous medium and the eluent at a first end **814** of each vessel **802** and to output a depleted stream, after removal of the element of interest in whole or in part, and a product stream, after desorption of the element of interest from the selective medium into the eluent, as effluents at a second end **816** of each vessel **802**, opposite from the first end **814**. In this way, each of the vessels **802** is operated in co-flow mode, with absorption and desorption flow being in the same direction. The aqueous material and the eluent are both provided to the first end **814** of each vessel **802A** and **802B**, and the depleted stream and product stream are both recovered at the second end **816** of each vessel **802A** and **802B**. In other cases, the vessels **802** can be configured to operate in counter-flow mode, where absorption and desorption flows are in opposite directions.

[0093] The system **800** has a second flow system **820** to control flow of the depleted stream and the eluent at the second end **816** of the vessels **802A** and **802B**. Like the flow system **806**, which in this case is a first flow system, the second flow system **820** has piping and valves for controlling disposition of the depleted stream and the product stream of the vessels **802A** and **802B**. The system **800** has a third pump **810C** to route the depleted stream of the first vessel **802A** and the second vessel **802B** to any suitable disposition. The system **800** also has a fourth pump **810D** to route the product stream of the first vessel **802A** and the second vessel **802B** to any suitable disposition. The controller **850** is also operatively coupled to the piping and valves of the second flow system **820** to control the second flow system **820** to use the third pump **810C** to pump the product stream of the first vessel **802A** while using the fourth pump **810D** to pump the depleted stream of the second vessel **802A**, and to use the third pump **810C** to pump the product stream of the second vessel **802B** while using the fourth pump **810D** to pump the depleted stream of the first vessel **802A**. As with the flow system **806**, which in this case is a first flow system, controller **850** can be configured to control the second flow system **820** to flow and route the depleted stream and the product stream independently. Thus, the controller **850** can be configured to stop flow of the depleted stream altogether while allowing flow of the product stream from vessel **802A**, vessel **802B**, or both, and to stop flow of the product stream altogether while allowing flow of the depleted stream from vessel **802A**, vessel **802B**, or both. The controller **850** is further configured to coordinate flow settings of the first and second flow systems **806** and **820** so that the first and second vessels **802A** and **802B** operate in absorption mode or desorption mode, where operating a vessel in absorption mode comprises routing the aqueous material to the vessel using the first pump **810A** and routing a depleted stream from the vessel using the third pump **810C**, and operating the vessel in desorption mode comprises routing the eluent to the vessel using the second pump **810B** and routing the product stream from the vessel using the fourth pump **810D**. Thus, the controller **850** is configured to operate the first and third pumps **810A** and **810C** together to place a vessel in absorption mode, and to operate the second and fourth pumps **810B** and **810D** together to place a vessel in desorption mode.

[0094] Thus, the fourth pump **810D** can be configured to pump a product stream from the first vessel **802A** or the second vessel **802B** and the third pump **810C** can be configured to pump the depleted stream from the first vessel **802A** or the second vessel **802B**, and the controller **850** can be further configured to route the product stream of the first vessel **802A** or the second vessel **802B** to the third pump **810C** by switching the third pump **810C** suction between the product stream of the first vessel **802A** and the product stream of the second vessel **802B** and to route the depleted stream of the first vessel **802** and the second vessel **802B** by switching the fourth pump **810D** suction between the depleted stream of the first vessel **802A** and the depleted stream of the second vessel **802B**.

[0095] The system **800** may have a third sensor **812C** and a fourth sensor **812D**. The third sensor **812C** may be coupled to a stream of the first vessel **802A** that is an opposite of the stream sensed

by the first sensor **812A**. That is, if the first sensor **812A** is used to sense input streams to the first vessel **802A**, the third sensor **812C** may be coupled to effluent streams of the first vessel **802A**, and vice versa. Thus, the first sensor **812A** may be coupled to the aqueous material input to the first vessel **802A** and the third sensor **812C** may be coupled to the depleted stream outlet from the first vessel **802A**. Alternately, the first sensor **812A** may be coupled to the eluent inlet of the first vessel **802A** and the third sensor **812C** may be coupled to the product stream outlet from the first vessel **802A**. Likewise, the fourth sensor **812D** may be coupled to a stream of the second vessel **802B** that is an opposite of the stream sense by the second sensor **812B**. That is, if the second sensor **812B** is used to sense input streams to the second vessel **802B**, the fourth sensor **812D** may be coupled to effluent streams of the second vessel **802B**, and vice versa. Thus, the second sensor **812B** may be coupled to the aqueous material input to the second vessel **802B** and the fourth sensor **812D** may be coupled to the depleted stream outlet from the second vessel **802B**. Alternately, the second sensor **812B** may be coupled to the eluent inlet of the second vessel **802B** and the fourth sensor **812D** may be coupled to the product stream outlet from the second vessel **802B**. The third and fourth sensors **812C** and **812D** may be the same kind of sensors as the first and second sensors **812A** and **812B**. That is, all the sensors **812** may be the same kind of sensor, which can be any of the types of sensors described above. Alternately, the sensors **812** may be different types of sensors, which is to say that any of the sensors **812A**, **812B**, **812C**, and **812D** may be a different type of sensor than any of the other sensors **812A-D**. It should be noted that any of the sensors **812A-D** can be a sensor station comprising a plurality of independent or interdependent sensors. Where any of the sensors **812A-D** is a sensor station comprising a plurality of sensors, multiple sensors can be used to measure TDS or a parameter relating to TDS in order to improve estimation of the actual state of the stream. The multiple sensors in one sensor station may be the same type of sensor, and may measure the same characteristic, or may be different types of sensors, or may measure different characteristics relating to TDS.

[0096] The sensors **812** of the system **800** can be used to monitor a process of the system **800** for effective performance. For example, readings of the sensors **812** can indicate changes in concentration of the element of interest in any or all streams of the system **800**. Changes in related streams, such as the aqueous material and the depleted stream, or such as the eluent and the product stream, can be compared to ascertain changes in performance of the selective medium in either vessel **802**. Thus, where a time change in concentration of the element of interest in the aqueous material that does not match a change in concentration of the element of interest in the depleted stream, the selective medium may have developed an unwanted condition, such as saturation, channeling, leakage, or breakthrough. Likewise, where a time change in concentration of the element of interest in the eluent and the product stream do not match, a similar unwanted condition may be thought to exist. Under such circumstances, a regeneration of the selective medium may be performed, the selective medium may be replaced, or other remedial action can be taken. The controller **850** can be configured to operate the flow systems **806** and **820** and the pumps **810** to accomplish automated remedial actions such as flushing or back-flushing. For example, the controller **850** can operate the flow systems to route the eluent to one or both vessels **802** to flush the selective medium. The sensors **812** can also be used to detect maximum loading and/or unloading points of the selective medium in a vessel to determine when to switch the operating mode of a vessel between absorption and desorption. Such switching can be automated by configuring the controller **850** to operate the flow systems **806** and **820** to switch modes automatically.

[0097] In some cases, the parameter related to TDS may be bed volume displacement. It can be useful to monitor the available fluid volume in the vessels **800** to aid in controlling the system **800**. The available fluid volume in a vessel can be ascertained by flowing an aqueous material comprising an element of interest through the vessel containing the selective medium at a known flow rate and measuring a change profile in concentration of the element of interest at the vessel

effluent. The rate of change in concentration of the element of interest at the vessel effluent can be related to the bed volume displacement within the vessel by simple algorithms. The measured bed volume displacement can be used for future measurements, especially where readings from sensors at inlet and outlet streams of the vessel are to be compared and the system **800** is to be controlled based on the comparison. For example, a change in the bed volume displacement can be readily ascertained by comparing inlet and outlet readings of the sensors **812**. Where bed volume displacement is seen to decrease, a remedial action may be ordered, or flow rate through the vessel may be reduced to maintain residence time. In other cases, as noted elsewhere herein, the sensors **812** can be used to indicate unwanted leakage of a material into a part of the process where that material should not be found. For example, where a vessel is operating in desorption mode, a sensor **812** at the vessel inlet, where eluent is being routed to the vessel, can be used to detect a sudden, or otherwise noticeable and unexpected, rise in the element of interest, which may indicate leakage of the aqueous material into the inlet of the vessel. Similar methods can be used for a vessel in adsorption mode, where a similar decrease in the element of interest can indicate leakage of eluent in the vessel inlet. In other cases, where a parameter related to TDS, measured by a sensor **812**, is trending but has not yet reached an endpoint, timing of the endpoint can be predicted and any remedial action or maintenance scheduled for a future date. In one specific example, where seals that isolate aqueous material or depleted stream from eluent or product stream are aging, sensors may detect a trend in concentration of the element of interest indicating increasing leakage. An endpoint can be predicted using such data and replacement of the seals can be scheduled for a future date.

[0098] FIG. **9** is a process flow diagram of a system **900**, according to another embodiment. Like the other systems described herein, the system **900** is a system for extracting an element of interest from an aqueous material using multiple vessels operated in a cyclic, permuted pattern. The system **900** has a plurality of vessels **902** (**902A**, **902B**, etc., through **902Z**), more than the two vessels **802** of the system **800**, that are operated in cyclic, permuted fashion by a controller **950**. The system **900** has a first flow system **906** and a second flow system **920**, like the system **800**, for routing inputs and effluents according to the mode of operation of the individual vessels. Where some vessels are in absorption mode, the controller **950** operates the first flow system **906** to route aqueous material to the vessel inlets and the second flow system **920** to recover the depleted stream from the vessel outlets of the vessels operating in absorption mode. Where other vessels are in desorption mode, the controller **950** operates the first flow system **906** to route eluent to the vessel inlets and the second flow system **920** to recover the product stream from the vessel outlets of the vessels operating in desorption mode. The vessels may also operate in other modes, such as flushing or waiting modes. The controller **950** controls the flow systems **906** and **920** to operate each vessel according to its appointed mode at all times, cycling the vessels between modes by starting, stopping, and switching flows of the different streams to change the mode of each vessel to accomplish an overall continuous extraction process to recover the element of interest from the aqueous medium. In this way, the controller **950** can be configured to operate the system **900** as a simulated moving bed system.

[0099] As with the system **900**, sensors can be coupled to inlet and/or outlet streams of the vessels **902** to monitor performance of each vessel **902** and of the overall process being performed by the system **900**, which may be a continuous extraction process that produces a product stream having a target concentration of the element of interest. The sensors can be used to detect processing endpoints of each vessel so that modes can be switched at preferred times and/or preferred parameters for the process (such as flow rates), and the sensors can be used to detect unwanted circumstances such as leaks and substandard performance of the selective medium in one or more of the vessels **902**. The controller **950** can be configured to detect such conditions and to take programmed action to permute and cycle the vessels through the various modes, or to take automated remedial action such as flushing or regenerating one of more of the vessels **902**.

[0100] FIG. **10A** is a graph **1000** showing data that can be obtained using sensors as described herein. The data of the graph **1000** shows performance of a vessel, as described herein, that has an extraction medium. This data shows a parameter related to TDS for tracking concentration of an element of interest in an aqueous medium provided to, or obtained from, the vessel. This data can be any of the types of data described herein, and the data shows change in the parameter as the vessel is operated in a loading mode **1002**, where the extraction medium is loaded with the element of interest, a displacement mode **1004**, where the fluid in the vessel having high concentration of the element of interest is displaced with a fluid, such as an eluent, having low or zero concentration of the element of interest, and a removal mode **1006**, where an eluent is used to remove the element of interest from the medium within the vessel. As can be seen from the graph **1000**, the parameter, for example conductivity, tracks at a high level during loading mode **1002** as the sensor detects an aqueous medium having high conductivity, and thus high concentration of the element of interest, decreases during displacement mode **1004**, as a fluid having low conductivity, and thus low concentration of the element of interest, is used to displace the fluid having high concentration, and tracks to a low value during removal mode **1006**, as the eluent having low conductivity, and thus low concentration of the element of interest, is used to remove the element of interest from the medium. This data is relatively smooth, indicating no aberrant departures from the expected behavior of the parameter during the various modes.

[0101] FIG. **10B** is a close-up view of a portion of the graph **1000**. The data shows slight variability at this scale, highlighting the resolution that can be achieved in the data using fast, quasi-continuous or continuous, sensors such as conductivity sensors, density sensors, and the like.

[0102] FIG. **11** is a graph **1100** showing similar data, but the data has aberrations **1102**. The aberrations **1102** indicate unwanted performance of the system. Here, two curves are shown representing data collected at different locations of the vessel. In this case, the curve having higher values represents data collected at a location where an aqueous medium having the element of interest is provided to the vessel for recovery of the element of interest. The curve having lower values represents data collected at a location where a depleted stream is withdrawn from the vessel after being depleted, to an extent, of the element of interest. Where the element of interest has been partially removed from the aqueous medium, the depleted stream has correspondingly lower signal, for example conductivity, than the aqueous medium. The aberrant signal **1102** is detected during a period where low concentration fluid, such as eluent, is being provided to the vessel. The large spike indicates that high concentration fluid has briefly leaked into the feed to the vessel and has been detected by the sensor. The leak is small, so the aberration is not particularly observed in the lower curve, since the small surge of dissolved ions is diluted into the fluid within the vessel and has no observable effect on the concentration of the element of interest in the fluid withdrawn from the vessel. Nonetheless, the leak can be detected in this way and remedial measures can be taken.

[0103] The data of the graph **1100** shows a small incidence of leaks. FIG. **12A** is a graph **1200** showing data similar to that of the graph **1100**, but with more indications of leaks. A controller, as described herein, can be configured to detect these spikes in the data and, for example, to count the spikes in a time interval. Such a controller can also be configured to integrate such data into a leakage volume per unit time. Such data can be used to determine when leakage has reached a point at which action, such as seal maintenance, replacement, or other action, must be taken to reduce or eliminate the leaks.

[0104] Accordingly, methods are described herein that include measuring a first conductivity of a first aqueous material using a conductivity sensor; performing an operation on the aqueous material to change a concentration of an element of interest of the aqueous material, for example lithium, manganese, nickel, and/or cobalt, and yield a second aqueous material; after performing the operation, measuring a second conductivity of the second aqueous material using a conductivity sensor; comparing the first conductivity or a first value of a variable derived from the first conductivity with the second conductivity, respectively a second value of the variable, derived from

the second conductivity; and modifying a parameter of the operation based on the comparison, for example by modifying a flow rate of the aqueous material. The methods can include determining a total dissolved solids content (TDS) of the aqueous material based on the conductivity measurement, wherein the variable derived from the conductivity is the TDS.

[0105] Performing the operation can include extracting the element of interest, using a direct extraction method, from a first stream to yield a depleted stream derived from the first stream and depleted of the element of interest, wherein the direct extraction method includes capturing the element of interest using a media selective of the element of interest, or an electrochemical process, wherein the first conductivity measurement is performed in the first stream and the second conductivity measurement is performed in the depleted stream. Performing the operation can include, additionally or instead, releasing the element of interest captured in a selective media in an eluent stream yielding a loaded eluent stream, wherein the first conductivity measurement is performed in the eluent stream and the second conductivity measurement is performed in the loaded eluent stream. Performing the operation can include, additionally or instead, removing an impurity distinct from the element of interest from a second stream, optionally the depleted stream or a derivative thereof, to yield a pure stream, wherein the first conductivity measurement is performed in the second stream and the second conductivity measurement is performed in the pure stream. Performing the operation can include, additionally or instead, concentrating a third stream, optionally the depleted stream, the pure stream or a derivative of one of the depleted or pure stream, to yield a concentrated stream, wherein the first conductivity measurement is performed in the third stream and the second conductivity measurement is performed in the concentrated stream.

[0106] The methods above can include performing a plurality of operations including at least a first operation upstream of a second operation, and can also include measuring a first conductivity of a first aqueous material using a conductivity sensor; performing the first operation on the first aqueous material to change a concentration of an element of interest and yield a second aqueous material; after performing the first operation, measuring a second conductivity of the second aqueous material using a conductivity sensor; comparing a first value of a variable derived from the first conductivity with a second value of the variable, derived from the second conductivity; modifying a parameter of the first operation based on the comparison of the first and second values; performing the second operation on an aqueous material derived from the second aqueous material to change a concentration of an element of interest and yield a third aqueous material; and, after performing the second operation, measuring a third conductivity of the third aqueous material using a conductivity sensor.

[0107] In the methods above, the aqueous material derived from the second aqueous material can be the second aqueous material, and the method can include comparing a third value of the variable derived from the third conductivity with the second value of the variable; and modifying a parameter of the second operation based on the comparison of the second and third values. In other cases, the aqueous material derived from the second aqueous material is a fourth aqueous material, and in such cases the method can include measuring a fourth conductivity of the fourth aqueous material, comparing the third value of the variable derived from the third conductivity with a fourth value of the variable derived from the fourth conductivity, and modifying a parameter of the second operation based on the comparison of the third and fourth values.

[0108] The first conductivity and second conductivity, optionally third and/or fourth conductivity, described above can be measured using one conductivity sensor, and the first and second conductivity, and optionally the third and fourth conductivity, can be measured over time, and the variable derived from the first and/or second conductivity is representative of a trend of the TDS over time. The methods can include iteratively varying the value of the operation parameter, and, for each iteration, measuring at least one of the first and second conductivity, and modifying the parameter based on the measured at least one of the first and second conductivity for each iteration. Further, a value of the parameter can be selected for which a target variable among the first and

second variable value reaches an extremum.

[0109] Other methods are described herein that include extracting an element of interest, for example lithium, manganese, nickel, and/or cobalt, from an aqueous material using a simulated moving bed process, wherein the simulated moving bed process includes cyclically permuting a plurality of vessels containing a selective media for extracting the element of interest between a set of stations, wherein each station of the set is fluidly connected to a fluid circulation pump for circulating a station fluid into the corresponding vessel, wherein the cyclic permutation is obtained via switching positions of one or more fluid circulation devices, and wherein extracting the element of interest includes controlling the position of the one or more fluid circulating devices so that a first vessel is fluidly connected to a loading fluid circulation pump for circulating the aqueous material in the first vessel, loading the element of interest in the selective media, and subsequently controlling the position of the fluid circulating devices so that the first vessel is fluidly connected to an unloading fluid circulation pump for circulating an eluent in the corresponding vessel unloading the element of interest from the selective media to the eluent, and wherein the method further includes, when the first vessel is in a first station of the set: iteratively varying a value of a process parameter of the simulated moving bed process, and, for each iteration, measuring a fluid parameter, such as a conductivity or density, relative to the TDS content of a first station fluid circulating in the first vessel, and determining a target value of the process parameter of the simulated moving bed process based on the iterative measurements.

[0110] In these methods, the process parameter can be the flow rate of the first station fluid. In some cases, the target value of the process parameter is a first target value of the process parameter and the methods include, when a second vessel is in the first station, iteratively varying a value of the process parameter, and, for each iteration, measuring the fluid parameter relative to the TDS content of a first station fluid circulating in the second vessel, and determining a second target value of the process parameter of the simulated moving bed process based on the iterative measurements. In some cases, the first target value of the process parameter is applied at each cycle when the first vessel is in the first station, and the second target value of the process parameter is optionally applied at each cycle when the second vessel is in the first station.

[0111] The fluid parameter above can be measured using a proxy sensor that is one of: a conductivity sensor, a Coriolis flowmeter or a pressure sensor. The proxy sensor can be located in the first, respectively second, vessel, for example at the inlet and/or the outlet of the first, respectively the second, vessels. The methods can include measuring the fluid parameter at several predetermined locations in relationship with the first, respectively the second, vessel and optionally determining a first, respectively second, target value of the process parameter based on said measurements.

[0112] In the above methods, the process parameter can be a first process parameter and the methods can include, when the first vessel is in a second station, iteratively varying a value of a second process parameter of the simulated moving bed process, and, for each iteration, measuring a fluid parameter relative to the TDS content of a second station fluid circulating in the first vessel, and determining a target value of the second process parameter of the simulated moving bed process based on the iterative measurements, which can be applied at each cycle when the first vessel is in the second station. Here, the first process parameter can be the flow rate of the first station fluid and the second process parameter can be the second station fluid. The target values of the first, respectively second, process parameter can be determined corresponding to the fluid parameter value reaching an extremum.

[0113] In these methods, the simulated moving bed process can include a plurality of stations including the first and a third station that are fluidly connected to a same fluid circulation pump, and, when the third vessel is in the third station, a fluid parameter relative to the TDS content of a third station fluid circulating in the third vessel can be measured for each iteration of the process parameter, and the target value of the process parameter can be determined based on the

measurements of the fluid parameter for the first and third station fluid. The vessels can be cyclically permuted in counter-current fashion relative to the flow of the station fluids. The fluid parameter can be measured continuously during the cyclic permutation of the first, respectively second or third, vessel. The simulated bed process herein can include four subsets of station, each fluidly connected to a respective circulation pump.

[0114] These methods can include determining a first target value of the process parameter when the first vessel is in the first station based on the fluid parameter measurements for the first station fluid at a first time, and determining a third target value of the first process parameter when the first vessel is in the first station based on the fluid parameter measurements for the first station fluid at a second time, and detecting a default in the simulated bed process based on a comparison of the first and third target values, wherein the default optionally includes a leak.

[0115] Also described herein are systems for extraction of an element of interest, such as lithium, manganese, nickel, and/or cobalt, from an aqueous material using a simulated moving bed system, which includes a plurality of vessels containing a selective media for extracting the element of interest, one or more of fluid circulation devices, each configured to switch between a plurality of positions, and a control system configured to control the fluid circulation devices so that the fluid circulation devices switch between a plurality of configurations, wherein each configuration corresponds to a cyclic permutation of the plurality of vessels between a plurality of stations, wherein each station of the plurality is fluidly connected to a fluid circulation pump for circulating a station fluid into the corresponding vessel, wherein at least a first station is fluidly connected to a first tank including the aqueous material and a second station is fluidly connected to a second tank including an eluent. The system can, for example, include four subsets of stations, each fluidly connected to a respective circulation pump. The control system can, for example, be configured to cyclically permute the vessels in a counter-current arrangement relative to the flow of the station fluids.

[0116] Thus, any of the systems described herein can be a simulated moving bed system comprising a plurality of vessels, the plurality of vessels including a first vessel and a second vessel, with flow systems configured to route flow among the vessels, and the controller such systems can be configured to control the flow systems to route the flow of the aqueous material and the eluent among the vessels in a cyclic, permuted pattern.

[0117] The system also includes one or more proxy sensors for measuring a fluid parameter associated to the first station fluid, at least when a first vessel is in the first station. Herein, the control system is configured to, when the first vessel is in the first station, iteratively vary a value of a process parameter of the simulated moving bed process, and, for each iteration, use the one or more proxy sensors to measure a fluid parameter relative to the TDS content of the first station fluid circulating in the first vessel, and to determine a target value of the process parameter of the simulated moving bed process based on the iterative measurements.

[0118] In the system above, the process parameter can be the flow rate of the first station fluid. The target value of the process parameter can be a first target value of the process parameter, and the control system can be configured so that when a second vessel is in the first station, the control system iteratively varies a value of the process parameter, for each iteration, uses the one or more proxy sensors to measure the fluid parameter for the first station fluid circulating in the second vessel, and determines a second target value of the process parameter based on the iterative measurements. The control system can also be configured to apply the first target value of the process parameter at each cycle when the first vessel is in the first station, and optionally, to apply the second target value of the process parameter at each cycle when the second vessel is in the first station.

[0119] In these systems, the fluid parameter can include a conductivity or a density, and the proxy sensor can be a conductivity sensor, a Coriolis flowmeter, and/or a pressure sensor. The proxy sensor can be located in the first, respectively second, vessel, and can be situated at the inlet and/or

the outlet of the first, respectively the at least second, vessels.

[0120] The control system can be configured to use the one or more proxy sensors to measure fluid parameter at several predetermined locations in relationship with the first, respectively the second, vessel and optionally determining a first, respectively second, target value of the process parameter based on said measurements. The process parameter can be a first process parameter and, when the first vessel is in a second station, the control system can be configured to iteratively vary a value of the second process parameter of the simulated moving bed process, for each iteration, use the one or more proxy sensors to measure a fluid parameter relative to the TDS content of a second station fluid circulating in the first vessel, and determine a target value of the second process parameter of the simulated moving bed process based on the iterative measurements. The control system can be configured to determine the target value of the first, respectively second, process parameter corresponding to the fluid parameter value reaching an extremum.

[0121] These systems can include a subset of stations including the first and a third station that are fluidly connected to a same fluid circulation pump. The control system can be configured to, when the third vessel is in the third station, use the one or more proxy sensors to measure a fluid parameter relative to the TDS content of a third station fluid circulating in the third vessel for each iteration of the process parameter, and determine the target value of the process parameter based on the measurements of the fluid parameter for the first and third station fluid. The control system can be configured to use the one or more proxy sensors to continuously measure the fluid parameter during the cyclic permutation of the first, respectively second or third, vessel. The control system can also be configured to determine a first target value of the first process parameter when the first vessel is in the first station based on the fluid parameter measurements for the first station fluid at a first time, determine a third target value of the first process parameter when the first vessel is in the first station at a second time based on the fluid parameter measurements for the first station fluid at a second time, and detect a default in the system based on the comparison of the first and third target values, wherein the default is optionally a leak.

[0122] The disclosure also relates to a system for extracting an element of interest from an aqueous material, the system comprising a first vessel containing a first selective medium to extract an element of interest from an aqueous material; a second vessel containing a second selective medium to extract the element of interest from the aqueous material; a source of the aqueous material fluidly couplable to the first vessel and the second vessel; an eluent source fluidly couplable to the first vessel and the second vessel; a first sensor to measure total dissolved solids (TDS), or a parameter related to TDS, at the first vessel; a flow system to direct the aqueous material to the first vessel or to the second vessel and to direct the eluent to the first vessel or to the second vessel; and a controller configured to control the flow system to route one of the aqueous material and eluent to the first vessel; and iteratively vary a flow rate through the first vessel, for each iteration control the first sensor to measure TDS, or the parameter related to TDS, at the first vessel, and determine a target value of the flow rate through the first vessel based on one or more of the measurements.

[0123] In an embodiment, the flow system is a first flow system, and the system further comprises a second flow system to route a first effluent of the first vessel and a second effluent of the second vessel. In some embodiments, the controller is further configured to control the second flow system to route a depleted stream from the first vessel while the aqueous material is routed to the first vessel and to route a product stream from the second vessel while the eluent is routed to the second vessel, and to route a product stream from the first vessel while the eluent is routed to the first vessel and a depleted stream from the second vessel while the aqueous material is routed to the second vessel.

[0124] The flow rate through the first vessel is the flow rate of the aqueous material when the controller controls the flow system to route the aqueous material to the first vessel. Alternatively or additionally, the flow rate through the first vessel may be the flow rate of the eluent when the

controller controls the flow system to route the eluent to the first vessel.

[0125] In some embodiments, the controller is further configured to adjust the flow rate through the first vessel to the target value of the flow rate through the first vessel.

[0126] In some embodiments, the systems further comprises a second sensor to measure TDS, or a parameter related to TDS, at the second vessel. The controller may be further configured to iteratively vary a flow rate through the second vessel, for each iteration control the second sensor to measure TDS, or the parameter related to TDS, at the second vessel, and determine a target value of the flow rate through the second vessel. In such embodiments, the controller may be further configured to adjust the flow rate through the first vessel to the target value of the flow rate through the first vessel, and to adjust the flow rate through the second vessel to the target value of the flow rate through the second vessel.

[0127] In some embodiments, the (first) sensor is located at an outlet of the first vessel. In some embodiments, the second sensor is located at an outlet of the second vessel.

[0128] In some embodiments, the system may include a (third) sensor to measure TDS, or a parameter related to TDS, that may be present irrespective of the presence of the second sensor. The (third) sensor located at an inlet of the first vessel. The system may include a fourth sensor to measure TDS, or a parameter related to TDS, the fourth sensor located at an inlet of the second vessel. The fourth sensor may be featured in the system with or without the third sensor.

[0129] In some embodiments, the controller is further configured to compare the measurements of two or more of the first sensor, the second sensor, the third sensor, and the fourth sensor and to adjust the flow rate through the first vessel, the flow rate through the second vessel, or both based on the comparison.

[0130] The disclosure also relates to methods, comprising extracting an element of interest from an aqueous material using at least a first vessel containing a first selective medium and a second vessel containing a second selective medium, wherein extracting the element of interest includes coupling a source of the aqueous material to the first vessel and to the second vessel to load the element of interest onto the first and second selective medium and coupling an eluent source to the first vessel and to the second vessel to unload the element of interest from the first and second selective medium. The method further includes routing a target fluid to the first vessel, the target fluid being one of the aqueous material and eluent; iteratively varying a flow rate of the target fluid to the first vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the first vessel for each iteration; and determining a target value of the flow rate of the target fluid through the first vessel based on the one or more of the measurements.

[0131] In such methods, the extracting the element of interest is performed using a simulated moving bed process. Such simulated moving bed process includes cyclically permuting the first and second vessels between a set of stations, to have a first subset of the set of stations fluidly connected to an aqueous material source and a second subset of the set of stations is fluidly connected to an eluent source. The cyclic permutation is obtained via switching positions of one or more fluid circulation devices. Extracting the element of interest includes controlling the position of the one or more fluid circulating devices so that the first vessel is fluidly connected to the aqueous source in a first station of the set of stations, and subsequently controlling the position of the fluid circulating devices so that the first vessel is fluidly connected to the eluent source in a second station of the set of stations. In such methods, the vessels are cyclically permuted counter-current relative to the flow of the aqueous material and eluent.

[0132] Such methods may include routing the target fluid to the second vessel, iteratively varying a flow rate of the target fluid to the second vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the second vessel for each iteration; and determining a target value of the flow rate of the target fluid through the second vessel based on the one or more of the measurements.

[0133] Such methods may include routing the target fluid to the second vessel, iteratively varying a

flow rate of the target fluid to the second vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the second vessel for each iteration; and determining a target value of the flow rate of the target fluid through the second vessel based on the one or more of the measurements. Such methods may further comprise flowing the target fluid through the first vessel at a target value of the flow rate of the target fluid through the first vessel when the first vessel is in the first station and flowing the target fluid through the second vessel at the target value of the flow rate of the target fluid through the second vessel when the second vessel is in the first station.

[0134] In some embodiments, the parameter related to TDS comprises a conductivity or a density.

[0135] In some embodiments, the TDS, or the parameter related to TDS, is measured at the inlet or the outlet of the first vessel, or both.

[0136] In some embodiments, the target fluid is the aqueous material, and the method further includes routing the eluent to the first vessel, iteratively varying the flow rate of the eluent to the first vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the first vessel for each iteration; and determining a target value of the flow rate of the eluent through the first vessel based on the one or more of the measurements.

[0137] In some embodiments, flowing the target fluid to the first vessel is performed using one or more fluid circulation devices wherein the method includes detecting a leak in the one or more fluid circulation devices based on the one or more measurements.

[0138] In general, any of the extraction systems described herein can be, or can include, a simulated moving bed system. Any of the sensors described herein can be a continuous, in-line sensor, such as a density sensor, an inertial flow sensor, a nuclear sensor, a pressure sensor, a temperature sensor, a conductivity sensor, or a combination thereof.

[0139] For all the systems described herein, the element of interest can be one or lithium, nickel, cobalt or copper. The parameter related to TDS is bed volume displacement, and the controller can be further configured to signal a leak based on any of the readings of the sensors.

[0140] While the foregoing is directed to embodiments of the present invention, other and further embodiments of the present disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

Claims

1. A system for extracting an element of interest from an aqueous material, the system comprising: a first vessel containing a first selective medium to extract an element of interest from an aqueous material; a second vessel containing a second selective medium to extract the element of interest from the aqueous material; a source of the aqueous material fluidly couplable to the first vessel and the second vessel; an eluent source fluidly couplable to the first vessel and the second vessel; a first sensor to measure total dissolved solids (TDS), or a parameter related to TDS, at the first vessel; a flow system to direct the aqueous material to the first vessel or to the second vessel and to direct the eluent to the first vessel or to the second vessel; and a controller configured to: control the flow system to route one of the aqueous material and eluent to the first vessel; and iteratively vary a flow rate through the first vessel, for each iteration control the first sensor to measure TDS, or the parameter related to TDS, at the first vessel, and determine a target value of the flow rate through the first vessel based on one or more of the measurements.

2. The system of claim 1, wherein the flow system is a first flow system, and further comprising a second flow system to route a first effluent of the first vessel and a second effluent of the second vessel.

3. The system of claim 2, wherein the controller is further configured to control the second flow system to route a depleted stream from the first vessel while the aqueous material is routed to the first vessel and to route a product stream from the second vessel while the eluent is routed to the second vessel, and to route a product stream from the first vessel while the eluent is routed to the

first vessel and a depleted stream from the second vessel while the aqueous material is routed to the second vessel.

4. The system of claim 1, wherein the flow rate through the first vessel is the flow rate of the aqueous material when the controller controls the flow system to route the aqueous material to the first vessel.

5. The system of claim 1, wherein the flow rate through the first vessel is the flow rate of the eluent when the controller controls the flow system to route the eluent to the first vessel.

6. The system of claim 1, wherein the controller is further configured to adjust the flow rate through the first vessel to the target value of the flow rate through the first vessel.

7. The system of claim 1, further comprising a second sensor to measure TDS, or a parameter related to TDS, at the second vessel, wherein the controller is further configured to iteratively vary a flow rate through the second vessel, for each iteration control the second sensor to measure TDS, or the parameter related to TDS, at the second vessel, and determine a target value of the flow rate through the second vessel.

8. The system of claim 7, wherein the controller is further configured to adjust the flow rate through the first vessel to the target value of the flow rate through the first vessel, and to adjust the flow rate through the second vessel to the target value of the flow rate through the second vessel.

9. The system of claim 7, wherein the first sensor is located at an outlet of the first vessel and the second sensor is located at an outlet of the second vessel.

10. The system of claim 9, further comprising a third sensor to measure TDS, or a parameter related to TDS, the third sensor located at an inlet of the first vessel, and/or a fourth sensor to measure TDS, or a parameter related to TDS, the fourth sensor located at an inlet of the second vessel.

11. The system of claim 10, wherein the controller is further configured to compare the measurements of two or more of the first sensor, the second sensor, the third sensor, and the fourth sensor and to adjust the flow rate through the first vessel, the flow rate through the second vessel, or both based on the comparison.

12. A method, comprising: extracting an element of interest from an aqueous material using at least a first vessel containing a first selective medium and a second vessel containing a second selective medium, wherein extracting the element of interest includes coupling a source of the aqueous material to the first vessel and to the second vessel to load the element of interest onto the first and second selective medium and coupling an eluent source to the first vessel and to the second vessel to unload the element of interest from the first and second selective medium, wherein the method further includes routing a target fluid to the first vessel, wherein the target fluid is one of the aqueous material and eluent; iteratively varying a flow rate of the target fluid to the first vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the first vessel for each iteration; and determining a target value of the flow rate of the target fluid through the first vessel based on the one or more of the measurements.

13. The method of claim 12, wherein the extracting the element of interest is performed using a simulated moving bed process, wherein the simulated moving bed process includes cyclically permuting the first and second vessels between a set of stations, wherein a first subset of the set of stations is fluidly connected to an aqueous material source and a second subset of the set of stations is fluidly connected to an eluent source, wherein the cyclic permutation is obtained via switching positions of one or more fluid circulation devices, and wherein extracting the element of interest includes controlling the position of the one or more fluid circulating devices so that the first vessel is fluidly connected to the aqueous source in a first station of the set of stations, and subsequently controlling the position of the fluid circulating devices so that the first vessel is fluidly connected to the eluent source in a second station of the set of stations.

14. The method of claim 12, further including: routing the target fluid to the second vessel, iteratively varying a flow rate of the target fluid to the second vessel, measuring total dissolved

solids (TDS), or a parameter related to TDS, at the second vessel for each iteration; and determining a target value of the flow rate of the target fluid through the second vessel based on the one or more of the measurements.

15. The method of claim 13, comprising: routing the target fluid to the second vessel, iteratively varying a flow rate of the target fluid to the second vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the second vessel for each iteration; and determining a target value of the flow rate of the target fluid through the second vessel based on the one or more of the measurements, the method further comprising flowing the target fluid through the first vessel at a target value of the flow rate of the target fluid through the first vessel when the first vessel is in the first station and flowing the target fluid through the second vessel at the target value of the flow rate of the target fluid through the second vessel when the second vessel is in the first station.

16. The method of claim 12, wherein the parameter related to TDS comprises a conductivity or a density.

17. The method of claim 12, wherein the TDS, or the parameter related to TDS, is measured at the inlet or the outlet of the first vessel, or both.

18. The method of claim 13, wherein the vessels are cyclically permuted counter-current relative to the flow of the aqueous material and eluent.

19. The method of claim 12, wherein the target fluid is the aqueous material, wherein the method further includes routing the eluent to the first vessel, iteratively varying the flow rate of the eluent to the first vessel, measuring total dissolved solids (TDS), or a parameter related to TDS, at the first vessel for each iteration; and determining a target value of the flow rate of the eluent through the first vessel based on the one or more of the measurements.

20. The method of claim 12, wherein flowing the target fluid to the first vessel is performed using one or more fluid circulation devices wherein the method includes detecting a leak in the one or more fluid circulation devices based on the one or more measurements.
